

JEE MAIN REVISION MASTER: CAPACITOR

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 167 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Capacitance & Dielectric Insertion	156 PYQs	Capacitance
Very High	Capacitor Networks & Energy Redistribution	19 PYQs	Circuits & Energy

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Capacitance & Dielectric Insertion	$C_0 = \frac{\epsilon_0 A}{d}$, $\quad C = \frac{K \epsilon_0 A}{d}$, $\quad C_{\text{slab}} = \frac{\epsilon_0 A}{d - t(1 - 1/K)}$, $\quad q_p = q \left(1 - \frac{1}{K}\right)$
Capacitor Networks & Energy Redistribution	$C_{\text{parallel}} = \sum C_i$, $\quad \frac{1}{C_{\text{series}}} = \sum \frac{1}{C_i}$, $\quad V_{\text{common}} = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$, $\quad \Delta U_{\text{loss}} = \frac{C_1 C_2}{2(C_1 + C_2)} (V_1 - V_2)^2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Capacitance & Dielectric Insertion (Very High Yield — 156 PYQs)

Core Principle: Parallel plate capacitance, dielectric polarization, and partial dielectric slab insertion.

■ **Caution / JEE Pitfall:** If battery remains **CONNECTED**: $V = \text{const}$, C increases by K $\implies Q$ increases by K . If battery is **DISCONNECTED**: $Q = \text{const}$ $\implies V$ decreases by $1/K$.

■ **Speed Trick:** Energy stored $U = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$. When dielectric is inserted with disconnected battery, U decreases because electrostatic attraction does work.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A sphere of capacitance 100 pF is charged to a potential of 100 V. Another identical uncharged metal sphere is brought in contact with the ..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "A parallel plate air capacitor has a capacitance C . When it is half filled as shown in figure with a dielectric constant $K=5$, the perc..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "A parallel plate capacitor has capacitance C , when there is vacuum within the parallel plates. A sheet having thickness $\frac{1}{3}r$..." [Ans: D]

3.2 Capacitor Networks & Energy Redistribution (Very High Yield — 19 PYQs)

Core Principle: Equivalent capacitance reduction, charge sharing between interconnected capacitors, and Joule heating loss.

■ **Caution / JEE Pitfall:** When connecting with opposite polarity, use numerator $(C_1 V_1 - C_2 V_2)$ for common potential.

■ **Speed Trick:** Energy loss during charge sharing $\Delta U_{\text{(loss)}}$ is completely independent of wire resistance; it always turns into heat and EM radiation.

Representative Source PYQs:

- **[2025 Morning Shift] Q#3 (MCQ):** "Two capacitors C_1 and C_2 are connected in parallel to a battery. Charge-time graph is shown below for the two capacitors. The ener..." **[Ans: N/A]**
- **[2025 Morning Shift] Q#3 (MCQ):** "An electron is made to enter symmetrically between two parallel and equally but oppositely charged metal plates, each of 10 cm length. The e..." **[Ans: N/A]**
- **[2024 Evening Shift] Q#3 (MCQ):** "A galvanometer (G) of 2Ω resistance is connected in the given circuit. The ratio of charge stored in C_1 and C_2 is :..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

A parallel plate air capacitor has a capacitance C . When it is half filled as shown in figure with a dielectric constant $K=5$, the percentage increase in the capacitance is _____.

- (A) 33.34
- (B) 66.67 **[Correct]**
- (C) 200
- (D) 400

Correct Answer: B

Solution Analysis: For a parallel plate capacitor with plate area A , separation d , and air (or vacuum) between the plates, the capacitance is given by : $C = \frac{\epsilon_0 A}{d}$ For a parallel plate capacitor with plate area A , separation d , and dielectric material with dielectric constant K between the plates, the capacitance is given by : $C' = \frac{K\epsilon_0 A}{d}$ When a dielectric slab of thickness $t = (d/2)$ and dielectric constant $K=5$ is inserted, the system is with two capacitors connected in series : Capacitor 1 (C_1) is filled with dielectric, thickness $d_1 = (d/2)$

Benchmark Problem 2: JEE Main 2024 (Online) 5th April Morning Shift (Q#4)

Three capacitors of capacitances $25\mu\text{F}$, $30\mu\text{F}$ and $45\mu\text{F}$ are connected in parallel to a supply of 100V . Energy stored in the above combination is E . When these capacitors are connected in series to the same supply, the stored energy is $(9/x)E$. The value of x is _____.

Correct Answer: 86

Solution Analysis: $E = \frac{1}{2}(25+30+45)(100)^2$ (i) Also, $(9/x)E = \frac{1}{2}\left(\frac{1}{(1/25)+(1/30)+(1/45)}\right)(100)^2$ (ii) From (i) and (ii) $x=86$